# ITA0328 - MOBILE COMPUTING

## ASSIGNMENT 1 – CO1

### Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township

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## PLANNING ASSUMPTIONS AND DESIGN BASIS

The assignment describes a 12 km² university township with academic buildings, hostels, residential colonies and a main road. The licensed spectrum provides 336 duplex voice channels after control-channel overhead. Subscriber forecasts are 1,000, 10,000, 20,000 and 100,000 or more. Busy-hour traffic is 0.025 Erlang per subscriber and the target call-blocking probability is no more than 2%.

A 120° three-sector option is considered for locations where interference is high. The main comparison uses a conventional hexagonal macrocell grid, fixed cluster reuse, sectorisation with reuse, and dynamic channel borrowing / hierarchical microcells.

### Chosen engineering assumptions

| Parameter | Adopted value | Reason |
|---|---|---|
| Total service area | 12 km² | Given in the assignment. |
| Available traffic channels, S | 336 | Given after control-channel overhead. |
| Busy-hour traffic per subscriber | 0.025 Erlang | Given. |
| Target grade of service | 2% blocking | Given. |
| Minimum SIR target | 12 dB | Digital-style threshold; reasonable balance between quality and reuse. |
| Path-loss exponent, n | 2.5 open / 3 mixed / 4 dense | Use the supplied range; the local value must be confirmed by drive testing. |
| Reference macrocell radius, R | 0.50 km | Planning starting point; adjusted to meet capacity and coverage. |
| Primary reuse cluster, N | 7 | Valid hexagonal cluster size; conventional planning starting point. |
| Sectoring | 120° in hotspot / interference-limited sites | Reduces dominant co-channel interferers, with some trunking loss. |
| Traffic model | Erlang-B, loss system | Calls blocked when all traffic channels are occupied; no queue. |
| Coverage model | Ideal regular hexagons | Real boundaries will be irregular due to terrain and buildings. |

Important limitation: the brief does not provide measured propagation maps, exact base-station costs, building penetration losses, or a measured traffic distribution.

Therefore, the numerical results below are a transparent planning example, not a final radio-network certification. The simulator and equations are intended to be rerun with field measurements.

**Traffic forecast**

| Subscribers | Total busy-hour traffic | Average traffic density over 12 km² |
|---|---|---|
| 1,000 | 25 Erlangs | 2.08 Erlangs/km² |
| 10,000 | 250 Erlangs | 20.83 Erlangs/km² |
| 20,000 | 500 Erlangs | 41.67 Erlangs/km² |
| 100,000 | 2,500 Erlangs | 208.33 Erlangs/km² |
| 25,000 event attendees | 1,250 Erlangs in 1 km² | 1,250 Erlangs/km² in stadium zone |

The stadium event traffic is 25,000 × 0.05 = 1,250 Erlangs. This is approximately six times the average density of the 100,000-subscriber township forecast and is concentrated in only 1 km².

## TASK 1 – DESIGN ALTERNATIVE FREQUENCY-PLANNING STRATEGIES (5 MARKS)

A fixed spectrum can be reused by separating cells in space, reducing interference through directional antennas, or assigning channels according to real-time demand. Three different strategies are investigated below.

**Strategy 1: Fixed cluster frequency reuse (iterative assignment)**

The service region is divided into hexagonal cells. A cluster contains N cells, and each cell in the cluster receives a unique channel group. The same group is reused in other clusters sufficiently far away. Valid hexagonal cluster sizes are $N = i^2 + ij + j^2$, where i and j are non-negative integers. Examples include 1, 3, 4, 7, 9, 12, 13 and 16.

With N = 7, the 336 traffic channels are divided into seven groups of 48 channels. Each cell receives one group, and the seven groups repeat across the grid. This is a spatial-reuse method. The radio access technology may implement the channel group using FDMA/TDMA resource units; in a modern OFDMA system, the equivalent resources are subcarriers/resource blocks scheduled in time and frequency.

Control channels carry access, paging, synchronisation and signalling information. Traffic channels carry the user voice/data session. The given 336-channel

value is already stated after control-channel overhead, so the calculations treat all 336 as traffic resources.

**Iterative procedure / pseudocode**

INPUT: total channels S=336, candidate (i,j), hexagonal cell grid
FOR each valid pair (i,j):
N = i*i + i*j + j*j
Q = sqrt(3*N)
estimate co-channel SIR using path-loss exponent n
reject N if SIR is below the chosen threshold
divide S into N channel groups (integer allocation)
assign one group to each cell in a reference cluster
repeat the cluster pattern over the complete hex grid
calculate channels/cell, Erlang capacity and site count
SELECT the feasible design with the best cost-capacity trade-off

Preprocessing: create a traffic-density map, obtain a propagation/drive-test survey, choose a hexagonal grid, identify candidate base-station locations and reserve control resources. The operator bears the survey, planning and site-acquisition cost; the university or property owner may share site access costs under a commercial agreement.

Justification: fixed reuse is easy to plan, explain and audit. It is a useful baseline and gives predictable interference, but it cannot adapt efficiently to uneven traffic.

**Strategy 2: Sectorised frequency reuse (spatial filtering)**

Each macrocell is divided into three 120° sectors using directional antennas. Each sector has its own antenna pattern and channel allocation. Directional radiation reduces the number of strong first-tier co-channel interferers seen by a user in a sector. In the idealised first-tier model, the dominant interferer count falls from six omnidirectional interferers to approximately two for 120° sectoring.

The design principle is spatial filtering. The same frequency group can be reused more aggressively, or the original cluster size can be retained to obtain a larger SIR margin. The channel pool is divided among sectors; for N = 7, 48 channels per cell can be allocated as 16 channels per sector in a simple equal split. Actual allocations may be 16/16/16 or adjusted to 17/16/15 if the system supports unequal sector loads.

FOR each cell:
divide the cell into three 120-degree sectors
estimate traffic and interference in each sector
assign a channel/resource group to each sector

```
    orient antennas toward their service sectors
    calculate SIR using directional antenna gains
    if sector load exceeds its channel capacity:
      rebalance resources or add a microcell
  REPEAT for all cells and tune antenna tilt/power
```

Preprocessing: traffic heat map by sector, antenna azimuth/tilt survey, propagation measurements and an interference scan. The operator bears radio-planning and antenna-upgrade costs. Sectorisation is justified where buildings, stadiums or main roads create directional traffic and interference.

**Strategy 3: Dynamic channel borrowing with hierarchical macro–microcells**

A macrocell provides continuous township coverage. When a cell becomes busy, it may borrow an unused channel from a neighbouring cell, provided the donor cell remains below its own blocking threshold and the borrowed channel does not create unacceptable co-channel interference. Microcells or temporary small cells can be added in local hotspots. This combines trunking efficiency with hierarchical coverage.

The design principle is traffic adaptation. A shared channel pool improves utilisation when traffic is uneven. A microcell layer shortens link distance and increases spatial capacity in dense areas. Handoff control is needed when a user moves between macro and micro layers.

```
  INITIALISE fixed channel pools and neighbour-cell list
  FOR each call request in cell c:
    if a free local channel exists:
      allocate local channel
    else:
      search neighbour cells for an eligible idle channel
      if a safe channel is available:
        borrow channel and record donor/borrower
      else:
        block the call
    monitor traffic and interference
  IF a hotspot remains overloaded for the forecast period:
    deploy or activate microcell / temporary small cell
    update handoff and power parameters
  RETURN blocking, utilisation, interference and cost
```

Preprocessing: time-varying traffic measurements, neighbour lists, channel compatibility rules, handoff zones, backhaul availability and candidate small-cell locations. The operator bears the controller, backhaul, site and optimisation costs. This strategy is justified because demand is uneven and the stadium event is temporary and predictable.

### Comparison of the three strategies

| Strategy | Main principle | How resources are shared | Main advantage | Main limitation |
|---|---|---|---|---|
| Fixed cluster reuse | Spatial reuse | Fixed channel groups repeat every N cells | Predictable and simple | Poor adaptation to uneven traffic |
| 120° sectorisation | Spatial filtering | Channels divided among directional sectors | Higher SIR / lower interference | Smaller channel pool per sector; antenna cost |
| Dynamic borrowing + macro/micro | Trunking and local densification | Idle channels borrowed; small cells added in hotspots | Adapts to changing load | More control complexity, handoffs and backhaul |

## TASK 2 – IDENTIFY THE GOVERNING DESIGN PARAMETERS (2 MARKS)

| Strategy | Dominant parameter | Why it governs the design | Trend as demand grows |
|---|---|---|---|
| Fixed cluster reuse | Cluster size N (and Q = D/R) | N controls channels per cell and reuse distance; larger N improves SIR but reduces channels per cell. | Keep N only as large as needed for SIR; add cells/reduce R as load rises. |
| Sectorised reuse | Number of sectors and effective interferer count $i_0$ | Sector angle and antenna pattern determine spatial filtering and SIR; channels per sector | Increase sector capacity selectively; do not split channels into sectors that are too |

| | | | |
|---|---|---|---|
| | | determine blocking. | small. |
| Dynamic borrowing / hierarchy | Available pool per cell and cell radius R | Borrowing depends on donor spare capacity; microcell radius determines spatial reuse and handoff frequency. | Increase borrowing efficiency first, then add microcells where sustained demand exceeds capacity. |

For fixed reuse, N is the central trade-off: channels per cell are approximately S/N, while reuse distance is $D = R\sqrt{(3N)}$. For sectorisation, the effective number of strong interferers is the key interference parameter, while channels per sector govern blocking. For dynamic borrowing, spare-channel availability and the overload ratio (offered traffic / available Erlang capacity) determine whether borrowing succeeds.

As subscribers increase from 1,000 to 100,000+, a fixed channel pool cannot be made larger. Capacity must be increased by improving spatial reuse, reducing cell radius, sectorising selected cells, and borrowing unused resources where possible.

# TASK 3 – DERIVE CAPACITY AND INTERFERENCE FUNCTIONS MATHEMATICALLY (6 MARKS)

## 3.1 Hexagonal cluster size and reuse distance

For a hexagonal frequency-reuse pattern, moving i cells along one chain and then turning 60° and moving j cells gives the cluster size:

$N = i^2 + ij + j^2$

The centre-to-centre co-channel reuse distance is:

$D = R\sqrt{(3N)}$, and reuse ratio $Q = D/R = \sqrt{(3N)}$

For N = 7, a valid pair is (i,j) = (2,1). Therefore $Q = \sqrt{21} = 4.583$ and $D = 4.583R$. With R = 0.50 km, D = 2.291 km.

## 3.2 Hexagonal cell area and number of cells

A regular hexagon with circumradius R has area $A_{cell} = (3\sqrt{3}/2)R^2 \approx 2.598R^2$. The equivalent apothem is $\sqrt{3}R/2$. With R = 0.50 km:

$A_{cell} = 2.598 \times (0.50)^2 = 0.6495$ km².

The ideal minimum number of cells for 12 km² is 12/0.6495 = 18.48, hence at least 19 cells. A real design requires extra edge coverage and may need more sites due to buildings and irregular boundaries.

### 3.3 Channels per cell and Erlang-B capacity

For S = 336 channels and cluster size N = 7:

$C_{cell} = S/N = 336/7 = 48$ traffic channels per cell.

The Erlang-B recursion is $B(0,A)=1$ and $B(m,A)=A \cdot B(m-1,A)/(m + A \cdot B(m-1,A))$, where m is the number of channels and A is offered traffic in Erlangs. Solving $B(48,A)=0.02$ gives $A \approx 38.39$ Erlangs per cell.

This means a 48-channel cell can carry approximately 38.39 Erlangs at 2% blocking under the Erlang-B assumptions. The maximum carried traffic is $A(1-B) \approx 37.62$ Erlangs per cell.

The approximate network capacity for M cells is $C_{total} \approx M \times 38.39$ Erlangs at the 2% blocking design point. For 19 coverage cells, this is approximately 729 Erlangs. For 74 cells, it is approximately 2,840 Erlangs.

### 3.4 Cell count required by the subscriber forecasts

| Subscribers | Traffic (Erlangs) | Cells required at 38.39 E/cell | Planning interpretation |
|---|---|---|---|
| 1,000 | 25 | 1 (coverage minimum is 19) | Coverage, not capacity, dominates. |
| 10,000 | 250 | 7 (coverage minimum is 19) | 19-cell baseline has sufficient ideal capacity. |
| 20,000 | 500 | 14 (coverage minimum is 19) | 19-cell baseline is still adequate in the ideal model. |
| 100,000 | 2,500 | 66 | Capacity dominates; plan about 74 cells at R=0.50 km for margin. |

At R = 0.50 km, 12 km² contains approximately 19 cells by ideal area, but the 100,000-subscriber case requires at least 66 cells at the Erlang-B design point. A 74-cell layout

gives about 2,840 Erlangs of ideal capacity, a margin of approximately 13.6% over 2,500 Erlangs.

### 3.5 Co-channel SIR derivation

Assume a mobile is served by its nearest base station at distance approximately R, and the first tier contains $i_0$ co-channel interferers at distance approximately D. With path loss proportional to distance raised to n, received power is proportional to d^(−n). If all interferers are equidistant and have equal transmit power:

$SIR \approx (R^{(-n)}) / (i_0 D^{(-n)}) = (D/R)^n / i_0 = (\sqrt{(3N)})^n / i_0$.

In decibels: $SIR_dB = 10 \log_{10}[(3N)^{(n/2)}/i_0]$. For an omnidirectional hexagonal layout, $i_0 = 6$.

| N | n | $i_0$ | Approx. SIR | Interpretation |
|---|---|---|---|---|
| 7 | 2.5 | 6 | 10.25 dB | Below the 12 dB target in open-campus propagation. |
| 7 | 3 | 6 | 12.05 dB | Just meets the 12 dB target. |
| 7 | 4 | 6 | 18.66 dB | Meets the target with margin; matches the assignment sanity check. |
| 7 | 2.5 | 2 | 15.02 dB | Idealised 120° sectoring improves SIR. |
| 7 | 4 | 2 | 23.43 dB | Strong idealised SIR; actual antenna side lobes reduce the gain. |
| 13 | 2.5 | 6 | 12.10 dB | Larger reuse cluster can meet 12 dB in open terrain. |

The six-interferer result for N=7 and n=4 is approximately 18.7 dB, matching the check stated in the assignment. Sectorisation is represented by reducing the effective dominant interferer count; real patterns require antenna-pattern measurements and may not achieve the ideal two-interferer case.

### 3.6 Cost–capacity relation

Let C_s be the one-time planning/survey cost, C_site the installed cost per site, C_op the operating cost per site per planning period, and M the number of cells/sites. Then:

C_total_cost = C_s + M(C_site + C_op).

With a 2% blocking target and m = floor(S/N) channels per cell, let A_B(m,0.02) be the Erlang-B offered traffic supported per cell. Then:

Network capacity K(N,M) = M × A_B(floor(S/N),0.02).

A simple cost per supported Erlang is Cost/Erlang = [C_s + M(C_site+C_op)] / K(N,M).

For sectorisation, use the sum of the three sector capacities, not automatically three times the unsectorised capacity. If 48 channels are divided into three sectors of 16 channels, each sector supports about 9.83 Erlangs at 2% blocking, so the cell supports about 29.49 Erlangs. This is lower than the 38.39 Erlangs of one pooled 48-channel group because splitting the pool reduces trunking efficiency. Sectorisation is worthwhile when the SIR improvement enables a smaller N, lower outage, or better reuse.

For dynamic borrowing, capacity is calculated from the local channels plus eligible borrowed channels. Its cost includes a controller, signalling/backhaul and additional coordination. The benefit is reduced blocking in a busy cell when neighbours have spare channels.

## TASK 4 – BEST, AVERAGE AND WORST-CASE BEHAVIOUR (4 MARKS)

### 4.1 Mobile position and SIR

Best case – near the serving base station: the desired signal is strong because the serving distance is small. If the nearest interferers remain at roughly the reuse distance, SIR improves. A mobile close to the base station generally has a larger desired-to-interference ratio.

Average case – around the middle of the cell: desired signal power is lower than near the base station. Interferer distances vary with the mobile's exact location, so the six-equidistant approximation is a useful first estimate but not exact.

Worst case – near a cell edge or corner: the serving base station is farther away, and one or more co-channel base stations may be relatively close. Desired power falls while interference can rise. The cell corner is often a critical location because it is far from the serving site and may be exposed to several neighbouring cells.

For example, if the desired link distance increases from R to 1.5R while the dominant interferer distance stays D, the desired received power is multiplied by (1/1.5)^n. The SIR therefore falls by $10n \log_{10}(1.5)$ dB: approximately 4.40 dB for n=2.5, 5.28 dB for n=3 and 7.04 dB for n=4. This illustrates why edge users are more vulnerable.

### 4.2 Traffic condition and blocking

Off-peak: offered traffic is low and most channels are idle, so blocking is close to zero. Average period: occupancy rises but spare channels remain. Busy hour: the network is designed so the blocking probability is no more than 2%. Event surge: the offered traffic may exceed the engineered capacity and blocking can increase sharply.

For 48 channels, the 2% Erlang-B operating point is approximately 38.39 Erlangs. At 25 Erlangs, blocking is well below 2%. At 50 Erlangs, the blocking probability is higher than the target. Therefore, a cell with 48 channels should not be planned to carry 50 Erlangs if the 2% target is mandatory.

### 4.3 Shadowing and path-loss exponent

Shadowing is a slow variation caused by buildings, trees and terrain. A building can attenuate the desired signal more than the interfering signal, causing local SIR holes. Shadowing may also block an interferer and improve SIR at another location. A log-normal shadowing term $X\sigma$ can be added to the received power in dB: $Pr(dBm) = Pt(dBm) + \text{antenna gains} - PL(d) + X\sigma$.

The path-loss exponent changes the distance sensitivity. For the same N=7 and six-interferer model, SIR is about 10.25 dB at n=2.5, 12.05 dB at n=3 and 18.66 dB at n=4. A higher n reduces distant interference more strongly, but it also causes the desired signal to decay faster with distance. Consequently, dense built-up areas can have better reuse isolation but poorer coverage and more shadowing.

**4.4 Best / average / worst summary**

| Case | Mobile / traffic condition | Expected result | Reason |
|---|---|---|---|
| Best SIR | Near serving site; low shadowing | High SIR | Strong desired signal and usually weaker relative interference. |
| Average SIR | Typical locations and propagation | Near model prediction | Mixed distances and shadowing average out. |
| Worst SIR | Cell edge/corner behind obstruction | Low SIR / outage risk | Weak desired link and potentially nearby co-channel interferers. |
| Best blocking | Off-peak traffic | Very low blocking | Many idle channels. |
| Design blocking | Busy hour at engineered load | About 2% | Erlang-B capacity selected at target GoS. |
| Worst blocking | Event surge / overloaded sector | Can exceed 2% substantially | Offered traffic exceeds available channels. |

## TASK 5 – SCALING AND SENSITIVITY ANALYSIS (3 MARKS)

The main scaling relationships are:

- Channels per cell: $m \approx S/N$; therefore, increasing N reduces channels per cell approximately in proportion to 1/N.
- Reuse distance: $D/R = \sqrt{(3N)}$; increasing N improves the ideal SIR as N^(n/2).
- SIR: SIR $\propto$ N^(n/2)/$i_0$ under the equal-distance first-tier approximation.
- Number of cells for fixed area: $M \approx A_total/(2.598R^2)$, so $M \propto 1/R^2$.
- If cell radius is halved, cell area falls to one quarter and approximately four times as many cells are needed to cover the same area.

As the population rises from 1,000 to 100,000, traffic increases from 25 to 2,500 Erlangs, a factor of 100. The original spectrum does not grow, so capacity must come from spatial reuse and denser infrastructure. A 19-cell coverage-only layout is sufficient

for early stages in the ideal model but not for 100,000 subscribers. About 66 cells are required at the 2% design point; 74 cells provide a modest margin.

As n rises from 2.5 to 4, the ideal SIR improves for a given reuse distance because distant interferers decay more rapidly. However, the serving signal also decays more rapidly with distance, making cell-edge coverage and indoor penetration more difficult. Thus, a higher n can support tighter reuse but may require more sites or indoor systems.

Break-even judgement: fixed reuse is the best low-complexity baseline at low and moderate demand. At sustained demand approaching the 100,000-subscriber forecast, cell densification becomes necessary. Dynamic borrowing is preferable when overload is local and neighbouring cells have spare resources. Microcells / small cells become preferable when demand is persistently concentrated in a hotspot. Sectorisation is preferable when interference is the main limitation and traffic is directionally concentrated.

## TASK 6 – FORMULATE THE RECURRENCE FOR REPEATED CELL SPLITTING (3 MARKS)

Let $M_0$ be the initial number of macro cells, $R_0$ the original radius, and $C_0$ the offered-traffic capacity per cell at the selected blocking target. In one uniform split, each cell is replaced by four smaller cells of radius $R_{k+1} = R_k/2$. The four-cell assumption is a common idealised planning approximation for a halved radius.

Uniform splitting recurrence:

$M_{k+1} = 4M_k, \quad R_{k+1} = R_k/2.$

If each smaller cell receives an equivalent channel pool through spatial reuse, the ideal aggregate capacity per original cell is multiplied by four:

$K_{k+1} = 4K_k, \quad K_0 = M_0C_0.$

The transmit power needed to preserve a comparable received power at half the distance follows the path-loss relation $P_r \propto P_tR^{\wedge}(-n)$. Thus $P_{t,k+1} = P_{t,k}/2^n$. For n=3, each split stage requires approximately one-eighth the transmit power per cell, under this simplified link-budget scaling.

Handoff load increases as cells become smaller. A simple mobility proxy is that the boundary-crossing rate is inversely proportional to characteristic cell radius, so $H_{k+1}$ is approximately $2H_k$ for the same user speed and geometry.

Selective splitting: if only a fraction $f_k$ of cells is overloaded, the number of cells becomes:

$M_{k+1} = M_k(1 + 3f_k)$.

If a fraction $f_k$ is split into four, the total count increases by three for every split parent. If capacity of an unsplit cell is $C_k$ and a split parent produces four children, aggregate ideal capacity is:

$K_{k+1} = K_k + 3f_kM_kC_k$,

assuming each of the four children supports the same traffic capacity $C_k$. The base case is $M_0$ cells, radius $R_0$ and total offered load $A_0$. The work/cost per round is proportional to the number of new sites, approximately $3f_kM_k$.

The recurrence is an idealised upper bound. Real systems cannot simply duplicate the full 336-channel allocation in every cell without controlling co-channel interference; reuse planning, backhaul and regulatory limits remain applicable.

## TASK 7 – SOLVE THE RECURRENCE AND INTERPRET THE RESULT (4 MARKS)

For uniform splitting, $M_k = 4M_{k-1}$. Repeated substitution gives $M_1=4M_0$, $M_2=4(4M_0)=4^2M_0$, and in general:

$M_k = M_04^k$.

If ideal capacity also quadruples per round, $K_k = K_04^k$. To support demand D, require $K_04^k \geq D$. Dividing by $K_0$ and taking logarithms:

$4^k \geq D/K_0 \Rightarrow k \geq \log_4(D/K_0)$.

The minimum integer number of rounds is $k = \text{ceil}[\log_4(D/K_0)]$ when $D > K_0$; otherwise k=0.

Example: with 19 initial cells and $C_0$ = 38.39 Erlangs/cell, $K_0$ = 729.4 Erlangs. For D=2,500 Erlangs, $D/K_0=3.43$. Therefore $k=\text{ceil}(\log_4 3.43)=1$. One ideal uniform split would produce 76 cells and theoretical capacity of about 2,918 Erlangs. A practical layout may use selective splitting to avoid upgrading every cell.

For selective splitting, if a constant fraction f of cells is split each round, $M_k = M_0(1+3f)^k$. If the aggregate capacity grows by the same factor under the equal-child-capacity assumption, then:

$k = \text{ceil}[\ \log(D/K_0) / \log(1+3f)\ ]$.

Example with f=0.25: the growth factor is 1.75. Reaching 2,500/729.4 = 3.43 times the initial capacity requires $k=\text{ceil}(\log(3.43)/\log(1.75))=3$ rounds. This is slower than uniform splitting but avoids unnecessary sites in lightly loaded areas.

Cost and mobility interpretation: uniform splitting grows the site count rapidly and raises installation/backhaul cost. Smaller radii reduce per-site transmit power but increase handoff frequency, neighbour relations and mobility-management signalling. Selective splitting is generally more economical when traffic is concentrated in a few buildings or stadium areas.

## TASK 8 – IMPLEMENT AND TEST THE PLANNING STRATEGIES (4 MARKS)

A reproducible Python model is provided in Appendix A. It includes: (1) valid cluster-size generation, (2) hexagonal cell area and reuse distance, (3) Erlang-B blocking and inverse capacity, (4) first-tier SIR calculation, (5) a simple fixed-reuse/sectorised comparison, and (6) a basic dynamic borrowing routine. The same traffic inputs can be run for all strategies.

Test cases required by the brief are N = 3, 4, 7 and 12; path-loss exponents n = 2.5, 3 and 4; sectorised and unsectorised layouts; and subscriber counts 1,000, 10,000, 20,000 and 100,000. Use a fixed random seed for Monte-Carlo comparisons so that each strategy sees the same generated mobile locations and traffic sequence.

Sanity check: for N=7, n=4 and six equidistant interferers, the model returns approximately 18.66 dB, which rounds to 18.7 dB as required.

The code is a planning-level simulator rather than a full standards-compliant LTE/5G simulator. It does not implement detailed OFDMA scheduling, fading, antenna patterns, protocol signalling, or a complete handoff state machine. These can be added after the basic analytical model is validated.

## TASK 9 – RECORD AND PRESENT EXPERIMENTAL PERFORMANCE (3 MARKS)

For each strategy and each demand level, record the following metrics:

- SIR distribution: mean SIR, median SIR, 5th-percentile SIR and percentage of users below the 12 dB threshold.
- Outage probability: fraction of simulated users whose SIR is below the selected threshold.
- Blocking probability: Erlang-B result for each cell/sector and observed fraction of rejected call requests in the event simulation.
- Capacity and utilisation: channels per cell, offered traffic, carried traffic and channel occupancy.

- Infrastructure: number of sites, sectors, microcells, and any temporary event sites.
- Cost and computation: survey cost, site cost, operating cost, controller/backhaul cost and simulator run-time.

Recommended plots: (1) SIR cumulative distribution functions for each strategy; (2) blocking probability versus offered load; (3) supported capacity versus cluster size N; (4) number of sites versus subscriber count; and (5) cost per supported Erlang versus subscriber count. Keep axes, units and random seeds identical so the comparison is fair.

| Metric | Why it matters | Expected comparison |
|---|---|---|
| 5th-percentile SIR | Shows weak-edge users rather than only the average | Sectorisation should improve the lower tail if antenna isolation is effective. |
| Blocking probability | Directly checks the 2% grade-of-service target | Borrowing helps uneven loads; microcells add capacity in hotspots. |
| Sites required | Measures densification | Cell splitting increases site count; selective splitting limits it. |
| Cost per Erlang | Combines capacity with planning/infrastructure expense | Dynamic methods cost more to operate but may avoid permanent sites. |
| Run-time | Shows algorithmic complexity | Fixed reuse is simplest; dynamic allocation needs more state and checks. |

## TASK 10 – COMPARE THEORY WITH SIMULATION (4 MARKS)

Analytical calculations and simulation should agree in trend, but exact values may differ. For the ideal N=7, n=4, six-interferer model, both should approach 18.7 dB when the mobile is at the assumed reference position and interferers are equidistant. If the simulation includes random mobile positions, the mean and 5th percentile will differ from this single-geometry value.

For blocking, Erlang-B assumes Poisson call arrivals, exponentially distributed holding times, an infinite-source population and no waiting queue. A finite user population, bursty event arrivals, call retries and time-varying traffic can produce different observed blocking.

Main causes of differences:

- Equidistant-interferer approximation: real interferer distances vary with mobile position and cell layout.
- Shadowing and fading: random propagation changes desired and interfering powers.
- Finite tiers: a finite simulated grid has edge cells and may omit distant interferers.
- Mobile-position distribution: uniform random users differ from a real campus traffic heat map.
- Erlang-B assumptions: real call arrivals and holding times may not be Poisson/exponential.
- Integer channel allocation: S/N may not be an integer, so some cells receive one more channel than others.
- Sector antenna patterns: side lobes and back lobes mean the effective interferer count is not always exactly two.

The expected conclusion is that fixed reuse gives the most predictable baseline; sectorisation improves interference-limited areas; dynamic borrowing handles local short-term load imbalance; and microcells provide the largest sustained local capacity increase but add cost and handoff complexity. The best design is therefore a hybrid, not simply the option with the highest theoretical capacity.

## TASK 11 – RECOMMEND AND DEFEND THE FINAL DESIGN (4 MARKS)

**Recommended township design**

Deploy a hybrid macrocell network using N=7 fixed reuse as the baseline, with a 12 dB minimum SIR planning target, 120° sectorisation at selected interference-limited or high-traffic locations, dynamic channel borrowing between compatible neighbouring cells, and microcells at persistent hotspots. Use $R \approx 0.50$ km as the initial macrocell radius, then densify to approximately 74 cell-equivalents for the 100,000-subscriber planning horizon.

At N=7, 336/7 = 48 channels per cell. The ideal Erlang-B capacity at 2% blocking is approximately 38.39 Erlangs per cell. A 74-cell-equivalent network therefore supports about 2,840 Erlangs, compared with the 2,500 Erlangs forecast for 100,000 subscribers. The ideal six-interferer SIR is 10.25 dB at n=2.5, 12.05 dB at n=3 and 18.66 dB at n=4. Since open-campus n=2.5 does not meet 12 dB with unsectorised N=7, use 120° sectoring or N=13 in those specific open areas after confirming propagation measurements.

A global N=13 plan would reduce the channel allocation to about 25–26 channels per cell and lower per-cell Erlang capacity. Therefore, selective sectorisation or local reuse changes are more efficient than forcing the whole township to use the larger cluster size. In practice, cluster boundaries and channel groups must be engineered to prevent interference between sectors and neighbouring clusters.

### Trade-offs

| Design choice | Benefit | Cost / limitation |
|---|---|---|
| N=7 macro reuse | 48 channels/cell; good capacity efficiency | Open-area SIR may be below 12 dB without sectoring. |
| 120° sectoring | Improves spatial isolation and SIR | Antenna and optimisation cost; channel pool fragmentation. |
| Dynamic borrowing | Uses spare channels in neighbouring cells | Coordination, signalling and interference checks required. |
| Microcells / cell splitting | Adds capacity exactly where traffic is concentrated | Site, backhaul, power and handoff cost. |
| 74 cell-equivalents at 100k | Ideal capacity margin above 2,500 E | Requires more sites than the 19-cell coverage-only layout. |

The design remains practical beyond 100,000 subscribers only if densification continues and the operator introduces additional layers such as indoor small cells, improved scheduling, data-specific capacity engineering and possibly additional spectrum when available. Under the assignment constraint that spectrum cannot increase, the long-term levers are smaller cells, better spatial reuse and more efficient radio resource management.

## TASK 12 – DAY-4 DESIGN CHALLENGE: STADIUM EVENT (8 MARKS)

### A. Impact analysis (2 marks)

The event brings 25,000 attendees into 1 km², each generating 0.05 Erlang. The event traffic is:

$A_{event} = 25{,}000 \times 0.05 = 1{,}250$ Erlangs.

Event traffic density = 1,250 Erlangs/km². For comparison, the 100,000-subscriber township average is 2,500/12 = 208.33 Erlangs/km². The stadium event density is about 1,250/208.33 = 6 times higher. Compared with the 20,000-subscriber stage, the township density is only 41.67 Erlangs/km², so the stadium event is 30 times that density.

The assumptions of uniform traffic, steady demand and permanent capacity no longer hold. The surge is concentrated in one zone and lasts four hours, and it occurs roughly twice a month. Its predictability and short duration favour a temporary or selectively activated capacity solution rather than permanently redesigning the entire township.

### B. Evaluation of the original design (2 marks)

A 1 km² stadium zone contains approximately 1/0.6495 = 1.54 macrocell areas at R=0.50 km, so it would be covered by roughly two or three macrocell footprints depending on boundaries. At 38.39 Erlangs per 48-channel cell, two cells provide only about 76.8 Erlangs and three cells about 115.2 Erlangs. The event offers 1,250 Erlangs, which is approximately 10.9 to 16.3 times this ideal capacity.

If the 1,250 Erlangs were evenly spread over three cells, each would receive about 416.7 Erlangs, far above the 38.39-Erlang design point. Blocking would be extremely high; the exact value depends on the actual number of channels and allocation, but the original design cannot meet the 2% target unchanged. The first failure is channel exhaustion and call blocking. If additional channels are forced into the zone without a reuse/antenna plan, co-channel interference can also become unacceptable.

### C. Adaptation or redesign (2 marks)

Decision: MODIFY the network temporarily on event days, while retaining the normal township macrocell layout. Use temporary small cells or cells-on-wheels (COWs) around the stadium, 120° sector antennas aimed into the stands and concourses, dynamic channel borrowing from lightly loaded neighbouring cells, and power/tilt optimisation. Add backhaul capacity and preconfigure handoff boundaries before the event.

At 48 channels per cell, one cell supports about 38.39 Erlangs at 2% blocking. The theoretical number of equivalent cells needed for 1,250 Erlangs is ceil(1,250/38.39) = 33 cells. This is a capacity-equivalent estimate, not a final site count: a real event network may use larger channel pools, multiple carriers/resource blocks, sectorisation,

and coordinated reuse. It demonstrates that two or three macro footprints are insufficient.

The temporary design should aim for about 33 cell-equivalents of 48-channel capacity, or an equivalent combination of sector capacities and available radio resources. Deploy only the number of temporary nodes that field testing and the detailed radio plan show is needed. Reduce transmit power to limit interference, coordinate reuse between adjacent temporary cells, and use make-before-break or robust handoff settings where supported. The additional nodes increase handoff events, so placement should avoid unnecessarily tiny cells.

**D. Evidence-based defence (2 marks)**

A permanent redesign of the whole township for an event occurring twice a month would create recurring capital, site-rental, power and maintenance costs for capacity that is idle most of the time. Retaining the existing network unchanged would fail the 2% blocking requirement because the stadium load is more than ten times the capacity of the nearby macrocell footprints. Temporary COWs/small cells, sectorisation and channel borrowing match the localised, predictable and short-lived nature of the demand.

The recommended choice is therefore MODIFY: retain the baseline macro network, temporarily densify the stadium zone for the four-hour event, and remove or deactivate temporary capacity afterwards. The decision balances capacity, SIR, handoff load and cost. A detailed site survey and rehearsal before the event are essential to validate the number and placement of temporary cells.

## CONCLUSION

The assignment shows that cellular capacity is determined not only by the total spectrum but also by the way channels are reused across space. Fixed cluster reuse provides a clear baseline, sectorisation improves interference isolation, and dynamic channel borrowing plus microcells adapts the network to uneven and changing demand. For the 12 km² township, a 19-cell coverage-only layout is sufficient at early subscriber levels in the ideal model, while the 100,000-subscriber forecast requires approximately 66 cells at the 2% Erlang-B design point. A 74-cell-equivalent plan provides a practical planning margin.

The N=7, n=4 first-tier SIR estimate is 18.7 dB, matching the assignment check. However, N=7 at n=2.5 gives only about 10.25 dB, so open-campus locations need sectorisation or a larger reuse cluster. For the stadium event, the 1,250-Erlang load is far

above the nearby macrocell capacity. Temporary small cells/COWs and sectorisation are the most economical response because the event is localised, predictable and lasts only four hours.

## APPENDIX A – PYTHON PLANNING SIMULATOR

The following Python 3 code uses only the standard library. It can be copied into a .py file or Google Colab cell. It calculates valid cluster sizes, cell geometry, Erlang-B blocking, the 2% capacity point and ideal first-tier SIR. The simple borrowing function demonstrates the allocation logic; a full radio simulator would require a detailed hex-grid and antenna model.

```
import math
import random

S = 336
TARGET_BLOCKING = 0.02
AREA_KM2 = 12.0
SUBSCRIBER_LEVELS = [1000, 10000, 20000, 100000]

def valid_cluster_sizes(limit=20):
    values = []
    for i in range(limit):
        for j in range(limit):
            n = i*i + i*j + j*j
            if 1 <= n <= limit and n not in values:
                values.append(n)
    return sorted(values)

def hex_cell_area(radius_km):
    return (3 * math.sqrt(3) / 2) * radius_km**2

def reuse_distance(radius_km, N):
    return radius_km * math.sqrt(3 * N)

def erlang_b(channels, offered_traffic):
    """Erlang-B blocking probability, stable recursive form."""
    b = 1.0
```

```
    for m in range(1, channels + 1):
        b = (offered_traffic * b) / (m + offered_traffic * b)
    return b

def erlang_capacity(channels, target=TARGET_BLOCKING):
    """Largest offered traffic whose Erlang-B blocking is <= target."""
    low, high = 0.0, max(1.0, channels * 3.0)
    for _ in range(100):
        mid = (low + high) / 2
        if erlang_b(channels, mid) > target:
            high = mid
        else:
            low = mid
    return (low + high) / 2

def sir_db(N, n, interferers=6):
    linear = (3 * N)**(n / 2.0) / interferers
    return 10 * math.log10(linear)

def channels_per_cell(total_channels, N):
    # Balanced integer allocation: cells receive floor(S/N) or ceil(S/N).
    return total_channels // N

def cells_for_coverage(area_km2, radius_km):
    return math.ceil(area_km2 / hex_cell_area(radius_km))

def cells_for_capacity(total_traffic, channels, target=TARGET_BLOCKING):
    cap = erlang_capacity(channels, target)
    return math.ceil(total_traffic / cap), cap

def borrow_channel(local_free, neighbour_free):
    """Return True if local or a safe neighbour channel can be allocated."""
    if local_free > 0:
        return True, "local"
    for idx, free in enumerate(neighbour_free):
        if free > 0:
            neighbour_free[idx] -= 1
```

```
            return True, "borrowed_from_neighbour_" + str(idx)
    return False, "blocked"

def report():
    print("Valid cluster sizes:", valid_cluster_sizes(16))
    print("\\nN=7, R=0.5 km")
    print("Reuse distance:", round(reuse_distance(0.5, 7), 3), "km")
    print("Cell area:", round(hex_cell_area(0.5), 4), "km^2")
    print("Coverage cells:", cells_for_coverage(AREA_KM2, 0.5))
    print("SIR N=7, n=4:", round(sir_db(7, 4), 2), "dB")
    print("SIR N=7, n=2.5:", round(sir_db(7, 2.5), 2), "dB")
    ch = channels_per_cell(S, 7)
    cap = erlang_capacity(ch)
    print("Channels/cell:", ch, "Erlang capacity:", round(cap, 2))
    for users in SUBSCRIBER_LEVELS:
        traffic = users * 0.025
        n_cells, _ = cells_for_capacity(traffic, ch)
        print(users, "users:", traffic, "Erlangs;",
              n_cells, "capacity cells")
    event_traffic = 25000 * 0.05
    print("Stadium event:", event_traffic, "Erlangs")
    print("Event capacity-equivalent cells:",
          cells_for_capacity(event_traffic, ch)[0])

if __name__ == "__main__":
    report()
```

## APPENDIX B – SAMPLE OUTPUT AND CHECKS

| Check | Expected result | Purpose |
|---|---|---|
| Valid cluster sizes | 1, 3, 4, 7, 9, 12, 13, 16 | Confirms hexagonal cluster relation. |
| N=7, R=0.5 km | D=2.291 km | Confirms reuse-distance calculation. |
| Hex cell area, R=0.5 km | 0.6495 km² | Confirms coverage geometry. |

| | | |
|---|---|---|
| N=7, n=4, six interferers | 18.66 dB ≈ 18.7 dB | Matches assignment sanity check. |
| 48 channels, 2% blocking | 38.39 Erlangs | Erlang-B capacity point. |
| Stadium traffic | 1,250 Erlangs | 25,000 attendees × 0.05 Erlang. |

For a complete experimental submission, run the code and paste the actual console output, then extend the model with a hexagonal coordinate generator and random mobile positions. Do not present fabricated Monte-Carlo results as measured results.

## REFERENCES AND MODELLING NOTES

- Assignment brief: ITA03-MP Assignment 1, Mobile Computing, CO1, Tasks 1–12 (uploaded PDF).
- Erlang-B loss formula and standard teletraffic interpretation: use as the analytical blocking model specified by the assignment.
- Hexagonal cellular geometry and frequency reuse: $N = i^2 + ij + j^2$; $D = R\sqrt{(3N)}$; $A_cell = (3\sqrt{3}/2)R^2$.

All numerical values are calculated from the assignment data and the stated modelling assumptions. Replace the assumed radius, path-loss exponent, antenna pattern and cost parameters with survey measurements before treating the design as a deployable network.